ADITYA GAJENDRAN

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SUMMARY

Software engineer with M.S. and B.S. degrees in Electrical and Computer Engineering and hands-on project experience developing full-stack applications, Python-based backend services, and database-driven systems. Skilled in building REST APIs, implementing secure and reliable software, and applying automated testing and CI/CD practices throughout the development process.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, MATLAB
Frontend: React, HTML, CSS
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs
Databases: PostgreSQL, SQLite, MongoDB
Testing & Code Quality: Pytest, pytest-cov, Vitest, Supertest, Ruff
DevOps & Tools: Git, GitHub Actions, Docker, Linux
Data & ML: pandas, NumPy, scikit-learn

PROJECTS

SupplyDesk — Full-Stack Inventory & Purchasing Management System | *September–October 2026*
React, TypeScript, Node.js, Express, PostgreSQL, Docker, Vitest, Supertest, GitHub Actions

- Built a full-stack inventory management application supporting product tracking, stock adjustments, supplier management, purchase orders, and partial stock receipts.
- Implemented REST APIs with PostgreSQL transactions and validation to prevent negative inventory, duplicate SKUs, and over-receiving.
- Developed session-based authentication with password hashing, CSRF protection, and role-based access control for administrators, managers, and read-only users.
- Added automated API integration tests and a GitHub Actions CI pipeline for frontend linting, production builds, database migrations, and backend tests.

OpsPulse — Website Uptime Monitoring Dashboard | *October 2026*
Python, FastAPI, SQLite, JavaScript, Pytest, GitHub Actions

- Built a local-first monitoring application with scheduled and on-demand website checks, SQLite history, and dashboard visualizations for uptime and response latency.
- Implemented FastAPI endpoints to manage monitors, trigger checks, retrieve historical results, and serve dashboard statistics.
- Added outbound-request safeguards including public-IP validation, IP-pinned HTTP/HTTPS connections, TLS hostname verification, and disabled automatic redirects to help mitigate SSRF risk.
- Configured GitHub Actions to run Pytest and Ruff across Python 3.12 and 3.13, with an enforced 85% minimum test-coverage threshold.

Web-Based Chess Training System | *Graduate Project | Jan–May 2026*
Python, JavaScript, Stockfish, machine learning, PGN data processing

- Built a local browser-based chess trainer with an interactive board, legal-move validation, and Stockfish-powered play.
- Implemented opening recognition using a transposition-aware opening book and a machine-learning classifier trained on PGN-derived board positions.
- Built engine-evaluated blunder detection from board/move features and a pipeline for PGN parsing, dataset generation, model evaluation, personalized opening recommendations, and automatic game logging.

Drone Video Monitoring Web Application | *Senior Design Project | Team of 4 | August 2024–May 2025*
HTML, CSS, JavaScript, MongoDB, Google Drive integration

- Owned end-to-end web-application development for a four-person senior design project while teammates developed the physical drone hardware.
- Built browser controls for live video streams from three drones, including drone selection and start/stop playback.
- Integrated MongoDB to retrieve the selected stream link; transferred recordings to Google Drive, organized footage by drone, and built an in-app library for playback.

EDUCATION

Binghamton University | M.S. Electrical and Computer Engineering | May 2026
Binghamton University | B.S. Electrical and Computer Engineering | May 2025

CERTIFICATIONS

Meta Full Stack Developer: Front-End & Back-End from Scratch | October 2026